

Polynomially bounded solutions are dense in the Jacobi spectrum

Closing the conjectured reverse inclusion in arXiv:1204.3322

candidate full solution—likely valid

1. Source question

Dale T. Smith studies the self-adjoint half-line difference equation

$$-\Delta(a_{n-1}\Delta y_{n-1}) + b_n y_n = \lambda y_n, \quad n = 0, 1, \dots, \quad (1)$$

where $a_{n-1} > 0$, $b_n \in \mathbb{R}$, and every solution obeys the left boundary condition $y_{-1} = 0$ [1]. Let B be the associated operator on $\ell^2(\mathbb{N}_0)$. Under the coefficient-growth and limit-point hypotheses displayed below, Smith's Theorem 2.3 proves

$$\overline{E} \subseteq \sigma(B), \quad E = \{\lambda \in \mathbb{R} : (1) \text{ has a polynomially bounded solution}\}.$$

for all $n > n_0$, then $\lambda \in \sigma(B)$.

Theorem 2.3. Suppose there are constants $C_1, \gamma, n_0, L > 0$ such that for all $n > n_0$

$$|a_{n-1}| \leq C_1 a_{n-1}, \quad \sum_{k=n_0+1}^n a_{k-1}^2 \leq L e^{\gamma n}$$

and suppose (1.1) is limit-point. Let

$$E = \{\lambda \in \mathbb{R} : \text{has a polynomially bounded solution}\}.$$

Then $\overline{E} \subset \sigma(B)$.

On PDF page 11, Theorem 3.2 records a polynomial bound for the orthogonal-polynomial solution at spectral-measure almost every λ , but the paper then says that more is needed to prove the conjecture $\overline{E} = \sigma(B)$.

But this last term is bounded independently in n . □

The following theorem is proved in Glazman [3], p. 197, Theorem 9.

Theorem 3.2. For any $\varepsilon > 0$ and for μ -almost all values of λ (where μ is the spectral measure of B), there is a constant $L(\varepsilon, \lambda) > 0$ such that for all n

$$|(-1)^n p(n; \lambda)| \leq L(\varepsilon, \lambda) n^{1/2+\varepsilon}.$$

Here the $p(n; \lambda)$ are the orthogonal polynomials obtained from (1.2).

But we need more than this to prove our conjecture that under the hypotheses of Theorem 2.1 $\overline{E} = \sigma(B)$.

The extra ingredient is that this particular spectral measure has support equal to the whole spectrum. That follows from cyclicity of the first Jacobi basis vector.

2. Exact statement

Write $c_n = b_n + a_n + a_{n-1}$. After the unitary sign change $y_n \mapsto (-1)^n y_n$, equation (1) becomes the Jacobi recurrence

$$a_n p_{n+1}(\lambda) + c_n p_n(\lambda) + a_{n-1} p_{n-1}(\lambda) = \lambda p_n(\lambda), \quad p_{-1} = 0, \quad p_0 = 1. \quad (2)$$

Let J be the corresponding Jacobi operator; it is unitarily equivalent to B and the sign change fixes the first basis vector e_0 . In the limit-point case J is self-adjoint. Its scalar spectral measure at e_0 is the orthogonality measure μ for the polynomials p_n .

Theorem 1 (Smith's conjecture). *Suppose that there are constants $C_1, \gamma, n_0, L > 0$ such that, for every $n > n_0$,*

$$|\Delta a_{n-1}| \leq C_1 a_{n-1}, \quad \sum_{k=n_0+1}^n a_{k-1}^2 \leq L e^{\gamma n}, \quad (3)$$

and suppose (1) is limit-point. Then

$$\boxed{\overline{E} = \sigma(B)}.$$

More precisely, for every $\varepsilon > 0$, the set of λ for which the canonical solution $(-1)^n p_n(\lambda)$ satisfies

$$|p_n(\lambda)| \leq C_{\lambda, \varepsilon} (n+1)^{1/2+\varepsilon} \quad (n \geq 0)$$

has full μ -measure and is dense in $\sigma(B)$.

3. Proof intuition

There are two logically independent halves. Smith's Shnol-type theorem says that any polynomially bounded solution forces its parameter into the spectrum; this is where the growth assumptions (3) enter.

For the converse density, orthonormality gives $\int |p_n|^2 d\mu = 1$. Weight these identities by the summable sequence $(n+1)^{-1-2\varepsilon}$ and apply Tonelli. For μ -almost every λ , all the values $p_n(\lambda)$ are then controlled by one finite weighted square sum, yielding the claimed polynomial bound. Finally, e_0 is cyclic: the three-term recurrence generates every basis vector from e_0 . Hence $\text{supp } \mu = \sigma(J)$. A full-measure subset of a measure is dense in its support, which supplies exactly the missing inclusion.

4. The two spectral-measure lemmas

Lemma 2 (cyclicity and full support). *Let J be any self-adjoint half-line Jacobi operator whose off-diagonal coefficients a_n are all nonzero, and let $\mu(S) = \langle E_J(S)e_0, e_0 \rangle$. Then e_0 is cyclic for J and*

$$\text{supp } \mu = \sigma(J).$$

Proof. Every finitely supported vector lies in the domain of the Jacobi expression. From

$$Je_n = a_n e_{n+1} + c_n e_n + a_{n-1} e_{n-1}$$

(with the e_{-1} term absent) and $a_n \neq 0$, induction gives a polynomial q_n such that $e_n = q_n(J)e_0$. Thus the cyclic subspace generated by e_0 contains every basis vector and is all of $\ell^2(\mathbb{N}_0)$.

The cyclic spectral theorem now represents J as multiplication by x on $L^2(\mu)$ and sends e_0 to the constant function 1. The spectrum of this multiplication operator is the closed topological support of μ . Therefore $\text{supp } \mu = \sigma(J)$. $\square$

Lemma 3 (almost-everywhere polynomial bound). *In the setting of Lemma 2, let p_n be the normalized recurrence polynomials, so that*

$$\int_{\mathbb{R}} |p_n(\lambda)|^2 d\mu(\lambda) = 1.$$

For each $\varepsilon > 0$, for μ -almost every λ there is a finite $C_{\lambda, \varepsilon}$ satisfying

$$|p_n(\lambda)| \leq C_{\lambda, \varepsilon} (n+1)^{1/2+\varepsilon} \quad (n \geq 0).$$

Proof. All summands are nonnegative, so Tonelli's theorem and orthonormality give

$$\begin{aligned} \int_{\mathbb{R}} \sum_{n=0}^{\infty} \frac{|p_n(\lambda)|^2}{(n+1)^{1+2\varepsilon}} d\mu(\lambda) &= \sum_{n=0}^{\infty} \frac{1}{(n+1)^{1+2\varepsilon}} \int_{\mathbb{R}} |p_n(\lambda)|^2 d\mu(\lambda) \\ &= \sum_{n=0}^{\infty} \frac{1}{(n+1)^{1+2\varepsilon}} < \infty. \end{aligned}$$

Consequently, for μ -almost every λ the pointwise series has a finite value $S_{\varepsilon}(\lambda)$. Each one of its terms is at most that value, and hence

$$|p_n(\lambda)| \leq \sqrt{S_{\varepsilon}(\lambda)} (n+1)^{1/2+\varepsilon}.$$

Take $C_{\lambda, \varepsilon} = \sqrt{S_{\varepsilon}(\lambda)}$. $\square$

5. Proof of Theorem 1

Fix one $\varepsilon > 0$ and let G_ε be the full- μ -measure set supplied by Lemma 3. For every $\lambda \in G_\varepsilon$, the sequence $(-1)^n p_n(\lambda)$ solves Smith's original equation (1) and is polynomially bounded. Thus

$$G_\varepsilon \subseteq E, \quad \mu(\mathbb{R} \setminus E) = 0.$$

We claim that $\sigma(B) \subseteq \overline{E}$. Since B and J are unitarily equivalent and e_0 is unchanged, Lemma 2 gives

$$\sigma(B) = \sigma(J) = \text{supp } \mu.$$

If $\lambda_0 \in \sigma(B)$ and U is any open neighborhood of λ_0 , then $\mu(U) > 0$ by the definition of support. Because E has full μ -measure, $\mu(U \cap E) = \mu(U) > 0$, so $U \cap E \neq \emptyset$. Hence $\lambda_0 \in \overline{E}$.

Conversely, Smith's Theorem 2.3 applies under (3) and the limit-point assumption and gives $\overline{E} \subseteq \sigma(B)$ [1]. Combining the two inclusions proves the equality. The same argument works for every fixed $\varepsilon > 0$, proving the final assertion.

6. Stronger density half and edge cases

Corollary 4 (growth-free converse-Shnol density). *For every self-adjoint half-line Jacobi operator with nonzero off-diagonal coefficients, and every $\varepsilon > 0$, the parameters whose canonical generalized eigenfunction is $O((n+1)^{1/2+\varepsilon})$ form a full-spectral-measure subset dense in the spectrum.*

This is precisely Lemmas 2 and 3; no coefficient-growth assumption is used.

Eigenvalues and singular spectrum. Atoms cause no problem, and no absolute continuity is used: every neighborhood of every support point has positive measure.

Unbounded spectrum. The density argument is local and uses only finite open neighborhoods.

Constants and boundary data. The constant $C_{\lambda,\varepsilon}$ may depend on λ ; no uniform bound is claimed. Also $p_{-1} = 0, p_0 = 1$, so $(-1)^n p_n$ is an admissible nonzero solution of the source equation.

Limit-circle scope. A chosen self-adjoint Jacobi extension can still have a cyclic scalar measure, but Smith's reverse inclusion is not available as stated. No equality for such extensions is claimed here.

7. Verification and novelty boundary

The only external mathematical input in the new inclusion is the standard cyclic spectral theorem; the cyclicity and Tonelli estimates are proved above. The other inclusion is exactly Smith's Theorem 2.3. The proof was stress-tested at the domain/cyclicity induction, identification of the orthogonality measure with the scalar spectral measure, support versus spectrum, the almost-everywhere quantifier, and passage from full measure to density.

A bounded novelty search used the run's exact-id and keyword indexes. It also covered the arXiv record and citation trail for 1204.3322, plus web searches combining the title, author, "polynomially bounded solution," "closure E ," Jacobi, Shnol, spectral measure, and generalized eigenfunction terminology. It found broad converse-Shnol and generalized-eigenfunction results, but no later paper explicitly resolving this exact conjecture. This is not a publication-level priority search.

There is an important provenance caveat. The downloadable TeX source of 1204.3322 contains a commented-out short proof sketch aimed at the same conclusion, although the rendered paper omits it and explicitly retains the statement as a conjecture. That sketch invokes positive spectral measure in a missing open part of the spectrum without recording why the chosen measure has full support. Lemma 2 supplies that bridge, while Lemma 3 makes the almost-everywhere estimate self-contained. Accordingly, confidence in validity is high, but novelty confidence is low.

Mathematical-validity confidence: 97/100. Recommended human review: verify the scalar spectral-measure identification and the source hypotheses of Theorem 2.3 first; then check whether the commented source sketch or an older Jacobi/converse-Shnol reference should receive priority credit.

References

- [1] D. T. Smith, *Spectral Analysis of a Class of Self-Adjoint Difference Equations*, arXiv:1204.3322 (2012).
- [2] G. Teschl, *Jacobi Operators and Completely Integrable Nonlinear Lattices*, Mathematical Surveys and Monographs 72, American Mathematical Society, 2000.